Stage 1: Reasoner fine-tuning

Embodied understanding & reasoning

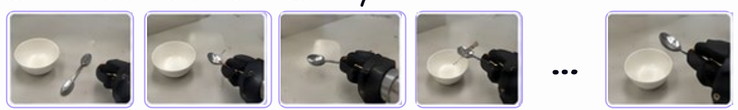
Spatial grounding

Next-subtask prediction

Structured video-prompt writing

Planning

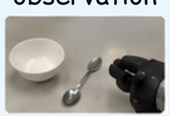
Task + observations + history



<subplan>
<world_modeling>
<duration>

Video-conditioned QA

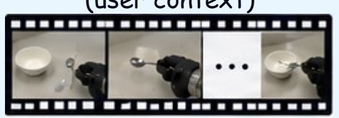
Question + observation



Video prompt (assistant)

The gripper moves the spoon into the bowl.

GT video (user context)



Answer (assistant)

A: The spoon is placed in the bowl.

Autoregressive loss on assistant tokens only

Stage 2: Generator fine-tuning

Freeze Reasoner

Self-forcing conditioning

1. Condition degradation

I2V start; multi-frame history where enabled

Selected visual conditions



Noise / optional blur



Degraded conditions

Offline-generated preceding clip



Precomputed once

2. Endpoint-only replacement

Last frame





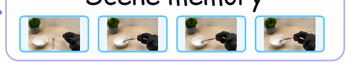
Replace current start

3. History + start replacement

Replace GT history source

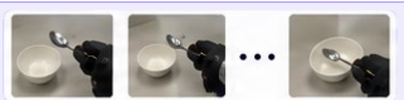
Same frame selection and encoding

Scene memory



Start frame / prefix

GT future supervision unchanged



Initial scene anchor retained